

Hyeonbin Han

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EDUCATION

Hanyang University ERICA

Expected Graduation Date: February 2027

GPA: 3.51/4.5

Department of Mathematical Data Science

EXPERIENCE

[CMES Robotics](#), Seoul, Korea

Mar 2026 – Jun 2026

- Junior Research Assistant, Robot Intelligence Team (Humanoid TF)
- Architected a scalable data acquisition and curation platform for the RB-Y1 humanoid robot, standardizing the teleoperation environment to ensure high-fidelity data collection.
- Deployed the Gello teleoperation framework for UR5 manipulators, streamlining the end-to-end pipeline from data curation to the training and physical deployment of Action Chunking Transformer (ACT) policies.

[Intelligence & Computation Lab](#), Hanyang University ERICA

Oct 2023 – Jan 2026

- Undergraduate researcher
- Administered and optimized the laboratory's GPU training infrastructure, ensuring high availability and efficient resource allocation for deep learning workloads.
- Engineered robust data preprocessing pipelines for heterogeneous healthcare datasets (including mobile addiction metrics), transforming raw signals into machine-learning-ready formats.

PUBLICATION

Han, Hyeonbin, et al. "Prediction of closed quotient during vocal phonation using GRU-type neural network with audio signals." (2024): 145-152.

CONFERENCE PRESENTATIONS

"DMD Method for Analyzing Interactions Between Board Game Players," KIICE Fall Conference, 2025 (Poster).

"Deep learning-based algorithm for analyzing EGG signals to predict closed quotient rate," KIIS Spring Conference, 2024 (Poster).

HONORS & AWARDS

Excellence Award, 2025 Intelligent Robot WE-Meet Project Integrated Competition, Jan 2026.

Participation Award, 2025 Shipbuilding & Maritime Big Data Utilization Idea Contest, Dec 2025.

Best Paper Award, 2025 KIICE Fall Conference, Oct 2025.

Bronze Prize (Robot News President's Award), KIICE Physical AI Challenge, Oct 2025.

Excellence Award, 2024 Hanyang University SW·ICT·AI Integrated Conference, Dec 2024.

SELECTED PROJECTS

Rethinking Adversarial Attacks and Defenses in Distributed Learning System

Sep 2025 – Jun 2026 (Ongoing)

- Joint capstone project with Information and Intelligent Security Lab, UNLV.
- Architected a reproducible Federated Learning (FL) pipeline using the Flower framework to simulate distributed training and fine-tune YOLOv8 on traffic signal datasets.
- Engineered Raspberry Pi-based edge vehicles, bridging software and physical hardware by deploying trained object detection models for real-time inference.
- Orchestrated physical security validations by injecting adversarial perturbations into the deployed model, empirically demonstrating vulnerabilities in autonomous environments.

Imitation Learning for Pick and Place Tasks

Sep 2025 – Dec 2025

- Industry-coupled problem-based learning (IC-PBL) project sponsored by CMES Robotics.
- Architected an end-to-end imitation learning pipeline by customizing and forking Hugging Face LeRobot and Robotis Physical AI tools to optimize industrial workflows.
- Streamlined high-fidelity data acquisition via teleoperation, curating a robust dataset of 200 demonstration episodes.
- Deployed state-of-the-art policies, including Action Chunking Transformer (ACT) and Diffusion Policy, directly onto the Robotis AI Worker FFW-SG2 platform for real-world execution.
- Validated model robustness and generalization capabilities by performing exhaustive testing across diverse lighting conditions and environmental variations.

Control Panel Anomaly Detection

Sep 2025 – Dec 2025

- Industry-coupled problem-based learning (IC-PBL) project sponsored by CANON Inc.
- Engineered a dual-stage vision pipeline integrating PatchCore for precise anomaly detection and EfficientNet-B0 for product classification.
- Optimized inference performance by extracting intermediate features (layers 4 and 6 of EfficientNet-B0) and accelerating K-Nearest Neighbors (KNN) computations through custom PyTorch tensor operations for GPU execution.
- Architected a comprehensive, no-code web interface using Streamlit to meet client requirements, streamlining the end-to-end workflow from data inspection, augmentation, and preprocessing to training and deployment.
- Implemented robust MLOps logging mechanisms to systematically track hardware status, terminal outputs, and experimental metrics, ensuring pipeline maintainability and reproducibility.

Box Depalletizing System Development

Mar 2025 – Jun 2025

- Industry-coupled problem-based learning (IC-PBL) project sponsored by CMES Robotics.
- Architected a robust 3D perception pipeline utilizing YOLOv8m on depth maps, optimizing for real-time inference while ensuring invariance to diverse box textures and environmental lighting.
- Engineered a scalable data augmentation workflow, bridging manual polygon annotations with synthetic depth data generated via Blender to resolve data scarcity and enhance model generalization.
- Formulated geometric transformations to compute surface normals from depth topography, extracting precise target coordinates and 6-DoF quaternion orientations for the end-effector.
- Orchestrated the seamless integration between the perception module and the RB5-850 collaborative robot, feeding the calculated spatial vectors into a teaching-based explicit path-planning algorithm.

Hand-held Camera Display State Classification

Mar 2025 – Jun 2025

- Industry-coupled problem-based learning (IC-PBL) project sponsored by CANON Inc.
- Engineered a lightweight vision pipeline utilizing MobileNetV3-Small, specifically optimizing the architecture for low-latency inference on resource-constrained edge devices.
- Formulated a targeted data augmentation strategy to account for real-world hand-held artifacts, effectively mitigating occlusion, motion blur, and sensor noise to maximize model generalization.
- Architected a secure, real-time inference environment via Cloudflare Tunnels (HTTPS), enabling seamless video stream ingestion directly from smartphone cameras to rigorously simulate actual field conditions.
- Deployed a responsive, mobile-first user interface tailored for on-site technicians, ensuring immediate field applicability and a frictionless end-to-end model serving experience.

TECHNICAL SKILLS

Languages: Python

AI & Machine Learning: PyTorch, Flower, Hugging Face LeRobot

Hardware & Fabrication: Raspberry Pi, Fusion 360, FreeCAD, 3D Printing

Infrastructure & Tools: Linux (Ubuntu), Cloudflare Tunnels, Docker, Streamlit, Git